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B1 AND B2

Abstract. abbbaaaaaaaa

1. Introduction

2. Model

Sun et al. [1] investigated a cancer growth model incorporating a strong Allee effect, tumor–immune interactions, and self- and cross-diffusion, with particular emphasis on its discrete space–time dynamics. We consider the spatially homogeneous continuous-time kinetics underlying their model. Writing

$$r = r_1, \qquad b = b_1, \qquad k = c_2, \qquad c = c_1, \qquad d = d_1$$

in the notation of [1], these kinetics take the form

$$\begin{aligned} \frac{dT}{dt} &= rT(1-bT)(T-m) - kTI, \\ \frac{dI}{dt} &= \frac{\rho TI}{\alpha + T} - cTI - dI. \end{aligned} \tag{1}$$

Here T and I denote the tumor and immune cell populations, respectively, and t denotes physical time. Throughout,

$$r, b, k, \rho, \alpha, c, d > 0, \qquad 0 < m < \frac{1}{b}. \tag{2}$$

The coefficient r sets the scale of tumor growth, b^{-1} is the tumor carrying capacity, and m is the strong Allee threshold. The parameter k measures immune-mediated tumor killing. In the immune equation, ρ is the maximal per-capita stimulation rate and α is the tumor population at half-maximal stimulation. The coefficients c and d describe tumor-induced immune inactivation and natural immune loss, respectively. No external immune-cell influx is included.

The biological meaning of the Allee assumption follows from the tumor per-capita growth function

$$g(T) = r(1-bT)(T-m).$$

In the absence of immune cells, this function is negative for $0 < T < m$, positive for $m < T < b^{-1}$, and negative for $T > b^{-1}$. Thus, the model assumes that sufficiently small tumor populations cannot sustain positive net growth. This is a phenomenological assumption about the tumor population under consideration, rather than a universal property of cancer growth. Moreover, m is the threshold for the immune-free subsystem; in the coupled system, tumor decline is determined by $g(T) - kI$.

The immune per-capita growth function is

$$h(T) = \frac{\rho T}{\alpha + T} - cT - d.$$

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Its stimulation component increases with tumor abundance and saturates, whereas the inactivation component grows linearly. Consequently, increased tumor abundance need not produce increased net immune growth. The variable I represents the immune population described by these assumptions, rather than the entire immune system: when $T = 0$, it satisfies $dI/dt = -dI$.

To obtain a polynomial vector field, introduce the time variable τ by

$$\frac{dt}{d\tau} = \alpha + T. \tag{3}$$

The chain rule transforms (1) into

$$\boxed{\begin{aligned} T' &= T(\alpha + T)\left[r(1 - bT)(T - m) - kI\right], \\ I' &= I\left[-cT^2 + (\rho - c\alpha - d)T - d\alpha\right], \end{aligned}} \tag{4}$$

where prime denotes differentiation with respect to τ. System (4) is an autonomous planar polynomial system of degree four. This transformation is exact and involves no approximation of the saturating response.

The biologically relevant phase space is

$$\mathcal{Q} = \{(T, I) \in \mathbb{R}^2 : T \geq 0,\ I \geq 0\}.$$

Both coordinate axes are invariant, and uniqueness of solutions implies that $\mathcal{Q}$ is positively invariant. Since $\alpha + T > 0$ on $\mathcal{Q}$, the original and polynomial systems are orbitally equivalent there, with the same time orientation. In particular, their equilibria, local stability types, periodic orbits, separatrices, and basins of attraction coincide.

The correspondence also preserves the forward asymptotic interpretation of trajectories. Indeed, the tumor equation gives

$$0 \leq T \leq M, \qquad M = \max\{T(0), b^{-1}\},$$

because the tumor component cannot increase above its carrying capacity. Hence, along each forward trajectory,

$$\alpha \leq \frac{dt}{d\tau} \leq \alpha + M.$$

The immune component has at most exponential growth on finite time intervals, so both formulations are forward complete, and $t \to \infty$ if and only if $\tau \to \infty$.

The rescaled time τ is an analytical variable. Physical time is recovered through

$$t(\tau) - t(0) = \int_0^\tau \left[\alpha + T(s)\right] ds. \tag{5}$$

Therefore, quantitative statements about growth rates, transient durations, or oscillation periods must be expressed in the original time variable.

We study (4) as a polynomial representation of the established tumor–immune kinetics. Its higher-degree terms arise from the common time rescaling and do not represent additional biological interactions. This formulation permits the use of polynomial dynamical-systems methods while retaining the biological interpretation of trajectories in $\mathcal{Q}$. The equivalence asserted here is restricted to this domain; an extension of the polynomial vector field outside $\mathcal{Q}$ requires separate mathematical interpretation.

References

[1] Ying Sun, Jinliang Wang, You Li, Yanhua Zhu, Haokun Tai, and Xiangyi Ma. Spatiotemporal complexity analysis of a discrete space-time cancer growth model with self-diffusion and cross-diffusion. *Advances in Continuous and Discrete Models*, 2024(1):37, 2024.

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